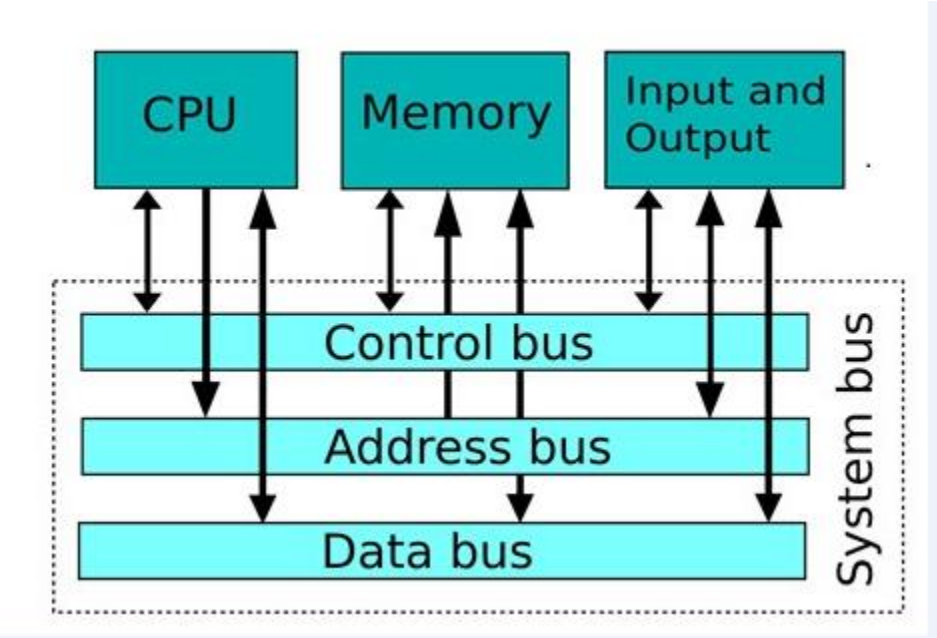
COMPUTER ORGANIZATION AND ARCHITECTURE

Computer organization refers to the components that make up a computer system.
Computer architecture is the science of integrating these computer components to achieve a level of functionality and performance.

Von Neuman Architecture

The Von Neuman architecture has got extensive application in everyday life. It consists of some important features and here we will have a detailed elaboration to the features.



CPU

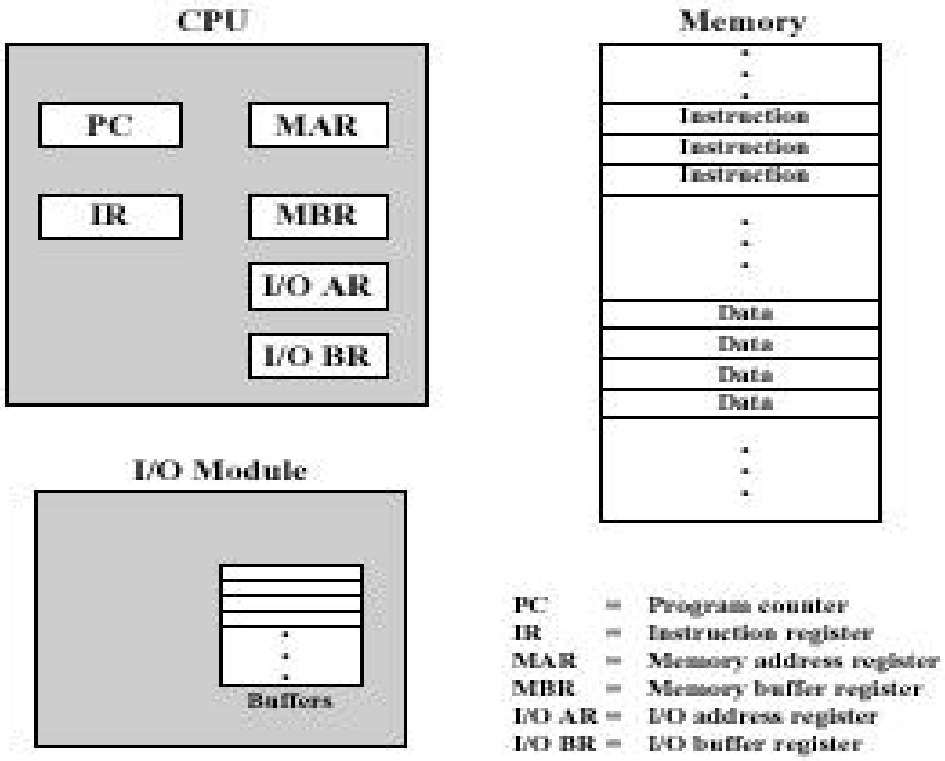
The **central processing unit (CPU)** of a computer is a piece of hardware that carries out the instructions of a computer program. Every instruction, no matter how simple, has to go through the CPU. It performs the basic arithmetical, logical, and input/output operations of a computer system. It is also simply referred to as the processor. All work that is done on a computer is performed directly or indirectly by the processor. The CPU is composed of three elements or parts: arithmetic logic unit (ALU), control unit and the registers.

i. **ALU:** the Arithmetic Logic Unit performs simple arithmetic and logical operations. It is involved in any sort of addition, subtraction, multiplication and division of data. In addition to that any other kind of algorithmic function and activities will be carried out by the ALU.

ii. **Control Unit:** it provides control signals in accordance with some timing which in turn controls the execution process. During the processing of data the control unit manages the data flow. The control unit follows the principle of the one at a time and accordingly it processes all the data. It decodes the program instructions. The control unit calls upon the arithmetic logic unit to perform the necessary calculations.

iii. **Registers**: fast stand-alone storage locations that hold data or instructions temporarily to speed up the operation of the processor. Multiple registers are needed to facilitate the operation of the CPU. They serve as high-speed memory where instructions can be copied to and retrieved. The following are the types of registers utilized by the processor during execution of instructions:

- Memory Address Register (MAR) – specifies address for next read or write instruction.
- Memory Buffer Register (MBR) – contains data to be written into or receives data read from memory.
- I/O address register - specifies a particular I/O device.
- I/O buffer register - used for exchange of data between an I/O module and CPU (or memory).
- Program Counter (PC) - keeps track of which instruction executes next.
- Instruction Register (IR)-holds the instruction that is currently being executed.



The processor plays a significant role in the following important aspects of a computer system;

Performance: while other components also play a key role in determining performance, the processor's capabilities dictate the maximum performance of a system. The other devices only allow the processor to reach its full potential.

Software Support: Newer, faster processors enable the use of the latest software. In addition, new processors, enable the use of specialized software not usable on earlier machines.

Reliability and Stability: The quality of the processor is one factor that determines how reliably your system will run. This also depends to some extent on the age of the processor and how much energy it consumes.

Energy Consumption and Cooling: Originally processors consumed relatively little power compared to other system devices. Newer processors can consume a great deal of power. Power consumption has an impact on everything from cooling method selection to overall system reliability.

Motherboard Support: The processor in use in your system will be a major determining factor in what sort of chipset we must use, and hence what motherboard you buy. The motherboard in turn dictates many facets of the system's capabilities and performance.

MEMORY

It is the other major subsystem in a computer. It consists of a collection of storage locations, each with a unique identifier, called an **address**. Data is transferred to and from memory in groups of bits called **words**. A word can be a group of 8bits, 16bits, 32bits or 64bits. If the word is 8bits, it is referred to as a **byte**. The term "byte" is so common in computer science that sometimes a16-bitword is referred to as a 2-byteword, or a 32-bit word is referred to as a 4-byteword. The table below shows the memory units in use.

Unit	*Exact Number of Bytes*	*Approximation*
kilobyte	2^{10} (1024) bytes	10^3 bytes
megabyte	2^{20} (1,048,576) bytes	10^6 bytes
gigabyte	2^{30} (1,073,741,824) bytes	10^9 bytes
terabyte	2^{40} bytes	10^{12} bytes

The memory stores data, results and programs to be executed. There are two classes of storage: Primary and Secondary.

Primary Storage

It consists of: RAM and ROM

i. RAM (Random Access Memory)

It is possible both to read data from the memory and to easily and rapidly write new data into the memory.

It is volatile - can only be used for temporary storage.

- Dynamic RAM- stores data as charge on capacitors which tend to discharge over time, therefore require periodic charge (like a memory reference) to refresh.
- Static RAM - stores data in traditional flip-flop logic gates. No refresh is needed and they are generally faster than dynamic RAMs.

ii. ROM (Read Only Memory) ROM is used to store data and program which is not going to change. It contains a permanent pattern of data which cannot be changed. Data is actually wired-in to the chip as part of the fabrication process.

However, there exist varieties of ROM that allow alteration of the contents of the chip:

- PROM - Programmable Read Only Memory

The writing process is performed electrically and may be done after chip fabrication. Once written, it cannot be erased.

- EPROM - Erasable Programmable Read Only Memory

It is read and written electrically. Before a write, all cells must be erased by exposure to UV radiation. Writing uses different electronics from normal memory writes. Errors can be corrected by erasing and starting over. They are more expensive than PROM.

- EEPROM - Electrically Erasable Programmable Read Only Memory

Any part(s) of the memory can be written at any time. It is updateable in place, that is, writing uses ordinary bus control, address, and data lines. Writing takes much longer than reading. It is more expensive (per bit) and less dense than EPROM.

- Flash Memory

This type of memory uses electrical erasing technology. It allows individual blocks to be erased, but not byte-level erasure, and modern flash memory is updateable in place (some may function more like I/O modules).

It has much faster erasure than EPROM.

It has the same density as EPROM.

Secondary Storage

Assignment Qn. 2: Describe the features of various types of secondary storage devices.

INPUT– OUTPUT

The input unit converts the external world data to a binary format, which can be understood by the CPU, e.g. mouse, joystick, keyboard etc.
The output unit converts the binary format data to a format that a user can understand e.g. monitor, printer, LCD, LED etc
Assignment Qn. 3: Describe five types of input devices and five output devices.

COMPUTER BUSES

This is a group of wires which carries information from the CPU to peripherals or vice versa. The CPU, memory and I/O devices are normally interconnected by the computer bus. A computer's main functions are data processing, data storage, data movement and control. Based on these, the computer buses have the following functions:

1. **Data sharing** - data refers to the actual digital pieces of information that need to be sent from one peripheral to another. All types of buses found on a computer must be able to transfer data between the computer peripherals connected to it. This data is transferred in either serial or parallel, which allows the exchange of 1, 2, 4 or even 8 bytes of data at a time. (A byte is a group of 8 bits). Buses are classified depending on how many bits they can move at the same time, which means that we have 8-bit, 16-bit, 32-bit or even 64-bit buses.

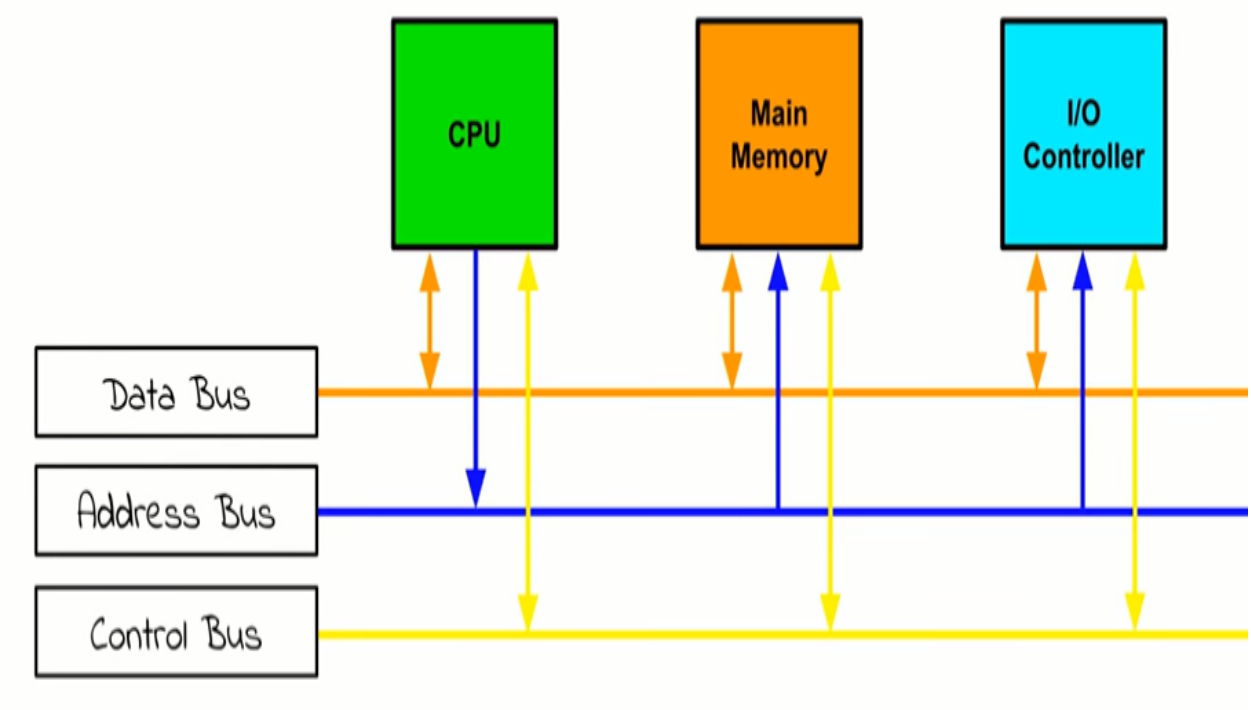
2. Addressing - A bus consists of address lines, which match those of the processor. This allows address information to be sent to or from specific memory locations. The address information describes where the data is located and where it needs to go during a particular operation.

3. Power - A bus supplies power to various peripherals that are connected to it.

4. Timing - The bus provides a system clock signal to synchronize the peripherals attached to it with the rest of the system. It transfers control signals that manage the flow of address and data. That includes which direction for the transfer of information and exactly how data needs to be routed through the computer system.

The expansion bus facilitates the easy connection of additional components and devices on a computer for example the addition of a TV card or sound card.

The bus structure is as shown in the figure below:



Data bus: this is a **bidirectional** bus that carries data bits from the processor to peripherals and vice versa.

Address bus: this is a **unidirectional** bus that carries address information bits from the processor to peripherals.

Control bus: it is a **bidirectional** bus that carries control signals from the processor to peripherals and vice versa.